Attention Is All You Need: Scalable Sequence Transduction via Self-Attention

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ABSTRACT

The dominant sequence transduction models are based on complex recurrent or convolutional neural networks that include an encoder and a decoder. The best performing models also connect the encoder and decoder through an attention mechanism. We propose a new simple network architecture, the Transformer, based solely on attention mechanisms, dispensing with recurrence and convolutions entirely. Experiments on two machine translation tasks show these models to be superior in quality while being more parallelizable and requiring significantly less time to train.

1. INTRODUCTION

Recurrent neural models, particularly long short-term memory (LSTM) and gated recurrent (GRU) neural networks, have been firmly established as state of the art approaches in sequence modeling and transduction problems such as language modeling and machine translation. Subsequent work has continued to push the boundaries of recurrent language models and encoder-decoder architectures. However, sequential computation remains a fundamental bottleneck in training efficiency.

2. MODEL ARCHITECTURE

The Transformer follows this overall architecture using stacked self-attention and point-wise, fully connected layers for both the encoder and decoder. The encoder is composed of a stack of N = 6 identical layers. Each layer has two sub-layers. The first is a multi-head self-attention mechanism, and the second is a simple, position-wise fully connected feed-forward network. Multi-Head Attention allows the model to jointly attend to information from different representation subspaces at different positions.

3. EXPERIMENTS & RESULTS

On the WMT 2014 English-to-German translation task, the big transformer model achieves a state-of-the-art BLEU score of 28.4, outperforming existing best models including ensembles by over 2.0 BLEU. On the WMT 2014 English-to-French translation task, our model establishes a new single-model state-of-the-art BLEU score of 41.8 after training for 3.5 days on eight GPUs.

4. RESEARCH GAPS & FUTURE WORK

While the Transformer demonstrates superior training speed and translation performance, long-context attention scales quadratically with sequence length O(N^2). Future research must address linear self-attention approximations, memory-efficient attention kernels, and cross-modal adaptation for multimodal scientific inputs.

5. CONCLUSION

In this work, we presented the Transformer, the first sequence transduction model based entirely on attention, replacing the recurrent layers most commonly used in encoder-decoder architectures with multi-headed self-attention.